Exercise 33.1

Sketch the variational free energy as a function of its one parameter $\bar{x}$ for a variety of values of the temperature T and the applied field h.

Solution:

Exercise 33.2

Show that, given $Q(\theta) = \delta(\theta - \theta^*)$, the optimal Q_k, in the sense of minimizing $\tilde{F}$, is a separable distribution in which the probability that $k_n = k$ is given by the responsibility $r_k^{(n)}$.

Solution:

Exercise 33.3

Show that, given a separable Q_k as described above, the optimal θ^*, in the sense of minimizing $\tilde{F}$, is obtained by the update step of the soft K-means algorithm. (Assume a uniform prior on θ.)

Solution:

Exercise 33.4

We can instantly improve on the infinitely large value of $\tilde{F}$ achieved by soft K-means clustering by allowing Q to be a more general distribution than a delta-function. Derive an update step in which Q is allowed to be a separable distribution, a product of $Q_\mu(\{\mu\})$, $Q_\sigma(\{\sigma\})$, and $Q_\pi(\pi)$. Discuss whether this generalized algorithm still suffers from soft K-means's 'kaboom' problem, where the algorithm glues an ever-shrinking Gaussian to one data point.

Solution:

Exercise 33.5

This exercise explores the assertion, made above, that the approximations given by variational free energy minimization always tend to be more compact than the true distribution. Consider a two-dimensional Gaussian distribution $P(x)$ with axes aligned with the directions $e^{(1)} = (1, 1)$ and $e^{(2)} = (1, -1)$. Let the variances in these two directions be σ_1^2 and σ_2^2. What is the optimal variance if this distribution is approximated by a spherical Gaussian with variance σ_Q^2, optimized by variational free energy minimization? If we instead optimized the objective function

$$G = \int dx\, P(x) \ln \frac{P(x)}{Q(x; \sigma^2)},$$

what would be the optimal value of σ^2? Sketch a contour of the true distribution $P(x)$ and the two approximating distributions in the case $\sigma_1/\sigma_2 = 10$.

Solution:

Exercise 33.6

What do you think of the idea of using a variational method to optimize an approximating distribution Q which we then use as a proposal density for importance sampling?

Solution:

Exercise 33.7

Define the relative entropy or Kullback-Leibler divergence between two probability distributions P and Q, and state Gibbs' inequality. Consider the problem of approximating a joint distribution $P(x, y)$ by a separable distribution $Q(x, y) = Q_X(x)Q_Y(y)$. Show that if the objective function for this approximation is

$$G(Q_X, Q_Y) = \sum_{x,y} P(x, y) \log_2 \frac{P(x, y)}{Q_X(x)Q_Y(y)},$$

then the minimum value of G is achieved when Q_X and Q_Y are equal to the marginal distributions over x and y.

Solution: